

Practical: Online Retail Dataset using HBase and HiveQL

Aim

To implement Online Retail analysis using HiveQL and HBase which includes:

1. Create and Load Online Retail table
 2. Create Index on table
 3. Find Total and Average Sales
 4. Find Order with Maximum Cost
 5. Find Customer with Maximum Order Total
 6. Find Country with Maximum and Minimum Sale
 7. Create External Hive Table connected to HBase
 8. Display Records from HBase Table
-

Step 1: Start Hadoop, Hive and HBase

```
start-dfs.sh
start-yarn.sh
start-hbase.sh
hive
```

Step 2: Create Database

```
CREATE DATABASE retaildb;

USE retaildb;
```

Step 3: Create Online Retail Table in Hive

```
CREATE TABLE online_retail(
  invoice_no STRING,
  product_name STRING,
  quantity INT,
  unit_price DOUBLE,
```

```
customer_id STRING,  
country STRING  
)  
ROW FORMAT DELIMITED  
FIELDS TERMINATED BY ','  
STORED AS TEXTFILE;
```

Step 4: Create online_retail.txt File

online_retail.txt

```
1001,Laptop,2,50000,501,India  
1002,Mobile,1,20000,502,USA  
1003,Headphone,4,2000,503,UK  
1004,Keyboard,3,1500,501,India  
1005,Mouse,5,800,504,Canada
```

Save file at:

```
/home/hadoop/online_retail.txt
```

Step 5: Load Data into Hive Table

```
LOAD DATA LOCAL INPATH '/home/hadoop/online_retail.txt'  
INTO TABLE online_retail;
```

Step 6: Display Records

```
SELECT * FROM online_retail;
```

Step 7: Create Index on Online Retail Table

```
CREATE INDEX idx_customer  
ON TABLE online_retail(customer_id)  
AS 'COMPACT'  
WITH DEFERRED REBUILD;
```

Rebuild Index

```
ALTER INDEX idx_customer  
ON online_retail  
REBUILD;
```

Step 8: Find Total Sales

```
SELECT SUM(quantity * unit_price) AS total_sales  
FROM online_retail;
```

Output

total_sales
132500

Step 9: Find Average Sales

```
SELECT AVG(quantity * unit_price) AS average_sales  
FROM online_retail;
```

Output

average_sales
26500

Step 10: Find Order Details with Maximum Cost

```
SELECT  
invoice_no,  
product_name,  
(quantity * unit_price) AS total_cost  
FROM online_retail  
ORDER BY total_cost DESC  
LIMIT 1;
```

Output

invoice_no	product_name	total_cost
------------	--------------	------------

1001	Laptop	100000
------	--------	--------

Step 11: Find Customer with Maximum Order Total

```
SELECT
customer_id,
SUM(quantity * unit_price) AS total_amount
FROM online_retail
GROUP BY customer_id
ORDER BY total_amount DESC
LIMIT 1;
```

Output

customer_id	total_amount
-------------	--------------

501	104500
-----	--------

Step 12: Find Country with Maximum Sale

```
SELECT
country,
SUM(quantity * unit_price) AS total_sales
FROM online_retail
GROUP BY country
ORDER BY total_sales DESC
LIMIT 1;
```

Output

country	total_sales
---------	-------------

India	104500
-------	--------

Step 13: Find Country with Minimum Sale

```
SELECT
country,
SUM(quantity * unit_price) AS total_sales
FROM online_retail
GROUP BY country
ORDER BY total_sales ASC
LIMIT 1;
```

Output

country	total_sales
Canada	4000

Step 14: Exit Hive and Open HBase Shell

```
exit;
```

```
hbase shell
```

Step 15: Create HBase Table for Online Retail

```
create 'online_retail_hbase',
'product_details',
'customer_details'
```

Step 16: Insert Records into HBase

```
put 'online_retail_hbase','1001','product_details:name','Laptop'
put 'online_retail_hbase','1001','product_details:price','50000'
put 'online_retail_hbase','1001','customer_details:id','501'
```

```
put 'online_retail_hbase','1002','product_details:name','Mobile'
put 'online_retail_hbase','1002','product_details:price','20000'
put 'online_retail_hbase','1002','customer_details:id','502'
```

Step 17: Display Records from HBase Table

```
scan 'online_retail_hbase'
```

Step 18: Create External Hive Table Connected to HBase

Exit HBase shell:

```
exit
```

Open Hive:

```
hive
```

Use Database:

```
USE retaildb;
```

Create External Hive Table

```
CREATE EXTERNAL TABLE retail_hive(  
    invoice_no STRING,  
    product_name STRING,  
    product_price STRING,  
    customer_id STRING  
)  
STORED BY 'org.apache.hadoop.hive.hbase.HBaseStorageHandler'  
WITH SERDEPROPERTIES (  
    "hbase.columns.mapping" =  
    ":key,  
    product_details:name,  
    product_details:price,  
    customer_details:id"  
)  
TBLPROPERTIES ("hbase.table.name" = "online_retail_hbase");
```

Step 19: Display External Hive Table Data

```
SELECT * FROM retail_hive;
```

Expected Output

invoice_no	product_name	product_price	customer_id
------------	--------------	---------------	-------------

1001	Laptop	50000	501
------	--------	-------	-----

1002	Mobile	20000	502
------	--------	-------	-----

Conclusion

Successfully implemented Online Retail analysis using HiveQL and HBase by:

- Creating and loading .txt data using comma delimiter
- Creating indexes
- Performing sales analysis
- Finding maximum order/customer sales
- Finding country-wise sales
- Connecting Hive with HBase using external table
- Displaying HBase records through HiveQL